Classification of quantum graphs over complex 3×3 matrices

(Klasyfikacja grafów kwantowych nad zespolonymi macierzami 3×3)

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Master's thesis

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Abstract

According to all known laws of aviation, there is no way a bee should be able to fly. Its wings are too small to get its fat little body off the ground. The bee, of course, flies anyway because bees don't care what humans think is impossible.

Według wszystkich znanych praw lotnictwa, pszczoła nie powinna w ogóle latać. Jej skrzydła są zbyt małe, by unieść jej pulchne ciałko z ziemi. Pszczoła, oczywiście, lata mimo wszystko, ponieważ pszczoły nie przejmują się tym, co ludzie uważają za niemożliwe.

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Chapter 1

Introduction

1.1 Motivation

Quantum graphs is a topic that has seen a resurgence in interest in recent years.

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I would like to thank my supervisor Mariusz Tobolski for his unwavering support and patience.

Chapter 2

Preliminaries

The following sections cover various topics in algebra. They are not intended to provide an exhaustive treatment; rather, they are meant to equip the Reader with the necessary background to understand the remainder of this work. References to further information are provided where appropriate.

2.1 Group theory

In the following section we will introduce some group theoretic notions that be used throughout this thesis. We assume that the Reader is familiar with the notion of a group. By $\mathbb{T}$ we will denote the torus, i.e. the set of complex numbers with the absolute value of 1, along with multiplication as its group action. By e_G we will denote the identity element of G. If G is abelian, its identity element will be expressed by 1_G.

Definition 2.1.1 (Dual of a group)**.** Let G be a group. The *dual group* $\hat{G}$ is the set of all homomorphisms from G into the torus $\mathbb{T}$ with the group operation defined as

$$(\varphi\psi)(g) = \varphi(g)\psi(g)$$

for all $\varphi, \psi \in \hat{G}$ and $g \in G$, where the latter multiplication takes place in $\mathbb{C}$.

Remark 2.1.1. We will often call the elements of the dual group *characters*. If a group is isomorphic to its complement it will be called *self-dual*.

The following proposition will be used throughout the thesis, as almost all of the groups we will be dealing with will be finite and abelian.

Proposition 2.1.1. *If G is finite and abelian, then it is self-dual.*

Proof. First we will prove the statement for finite cyclic groups. Let $G = \langle g \rangle$, $\mathrm{ord}(g) = n$. We will prove that the dual is cyclic of order n. Let $\chi\colon G \to \mathbb{T}$

be a character. Since the group is cyclic, the character is entirely determined by the value it takes on the generator g. Since $g^n = e$, obviously $\chi(g)^n = 1$, so $\chi(g) \in \mu_n = \{\exp(2k\pi i/n) : k = 0, 1, \ldots, n-1\} \subseteq \mathbb{T}$. For each possible choice of $\chi(g) \in \mu_n$ we get a different character. It is then easy to see that $\chi : G \to \mathbb{T}$ such that $\chi(g) = \exp(2\pi i/n)$ generates this group, i.e. $\hat{G} = \left\{\chi, \chi^2, \ldots, \chi^{n-1}, \chi^n = e_{\hat{G}}\right\}$.

Now since every finite abelian group can be written as a finite product of finite cycles, all we need to do is check whether the statement holds for a product of two groups A, B that are themselves self-dual, i.e. $\widehat{A \times B} \cong \widehat{A} \times \widehat{B}$. This is self-evident as every character $\chi \in \widehat{A \times B}$ is determined by its restrictions to $A \times \{e_B\}$ and $\{e_A\} \times B$, since

$$\chi(a, b) = \chi((a, e_B)(e_A, b)) = \chi(a, e_B)\chi(e_A, b).$$

Thus $\chi \leftrightarrow (\chi|_A, \chi|_B)$ is an isomorphism. $\square$

2.2 C^*-algebras

In the following section we will introduce some notions stemming from the theory of C^*-algebras. We assume the Reader is familiar with the notion of a C^*-algebra. For reference, see Blackadar [2].

Multiplication in a C^*-algebra A is a bilinear map

$$m\colon A \times A \to A.$$

By the universal property of the tensor product, m extends uniquely to a linear map

$$m^*\colon A \otimes A \to A.$$

Throughout this thesis, we will use the same notation (m) for both the bilinear multiplication and its unique linear extension, relying on the context to distinguish between the two.

Definition 2.2.1 (Linear functional)**.** A *linear functional* on a C^*-algebra A is a linear function $\psi : A \to \mathbb{C}$.

Remark 2.2.1. We will call linear functionals simply as *functionals*.

The following notions will be useful for defining an inner product on C^*-algebras. Positiveness and faithfulness are all we need to define a norm on a finite-dimensional C^*-algebra equipped with a suitable functional.

Definition 2.2.2 (Positive functional)**.** A functional ψ on a C^*-algebra A is called *positive*, if

$$0 \leq \psi(a^*a)$$

for any $a \in A$.

Definition 2.2.3 (Faithful functional)**.** A positive functional ψ on a C^*-algebra A is called *faithful*, if

$$\psi(a^*a) = 0 \Leftrightarrow a = 0$$

for any $a \in A$.

Remark 2.2.2. In the literature, the functional ψ is often called a *state*.

Example 2.2.1 (Positive and faithful functionals on matrix algebras)*.* Let $A = M_n(\mathbb{C})$, the C^*-algebra of $n \times n$ complex matrices, equipped with the usual trace Tr. Define

$$\psi(a) = n\,\mathrm{Tr}(a), \qquad a \in A.$$

Since a^*a is positive semidefinite for every $a \in A$, its eigenvalues are non-negative, and hence $\mathrm{Tr}(a^*a) \geq 0$; thus ψ is a positive functional. Moreover, $\mathrm{Tr}(a^*a) = 0$ forces all eigenvalues of a^*a to vanish, which (since a^*a is positive semidefinite, hence diagonalizable with non-negative eigenvalues) implies $a^*a = 0$ and therefore $a = 0$. So ψ is faithful.

Example 2.2.2 (A positive functional that is not faithful)*.* Let $A = \mathbb{C}^2 = \mathbb{C} \oplus \mathbb{C}$, with coordinatewise multiplication and involution given by complex conjugation in each coordinate. Define

$$\psi(x, y) = x, \qquad (x, y) \in A.$$

For $(x, y) \in A$ we have $(x, y)^*(x, y) = (|x|^2, |y|^2)$, so $\psi\big((x, y)^*(x, y)\big) = |x|^2 \geq 0$, and ψ is positive. However $\psi\big((0, y)^*(0, y)\big) = 0$ for every $y \in \mathbb{C}$, while $(0, y) \neq 0$ whenever $y \neq 0$. Hence ψ is positive but *not* faithful.

The following notion is akin to the notion of a group action on a set.

Definition 2.2.4 (Group action on a C^*-algebra)**.** By *group action* of a group G on a C^*-algebra A we mean a group homomorphism

$$\alpha\colon G \to \mathrm{Aut}(A).$$

We will write $g \cdot a$ instead of $\alpha(g)(a)$ then the action in question is clear.

Example 2.2.3 (Group actions on C^*-algebras)*.*

1. *Conjugation action.* Let $A = M_n(\mathbb{C})$ and let $G = U(n)$ be the group of unitary matrices. Then G acts on A via

   $$\alpha(u)(a) = uau^*, \qquad u \in U(n),\ a \in A.$$

   Each $\alpha(u)$ is easily checked to be a $*$-automorphism of A, and $\alpha(u_1u_2) = \alpha(u_1)\alpha(u_2)$, so $\alpha\colon G \to \mathrm{Aut}(A)$ is indeed a group homomorphism.

2. *Translation action.* Let $A = C(\mathbb{T})$, the continuous complex-valued functions on the torus $\mathbb{T}$, and let $G = \mathbb{Z}$ act by rotation:

$$(n \cdot f)(z) = f(z - n\theta \bmod 2\pi), \qquad n \in \mathbb{Z},\ f \in A,$$

for a fixed $\theta \in \mathbb{R}$. This is precisely the action used to construct the irrational rotation algebra as a crossed product.

3. *Trivial action.* For *any* C^*-algebra A and *any* group G, setting $\mathrm{triv}(g)(a) = a$ for all $g \in G,\ a \in A$ trivially defines a group action, since $\mathrm{triv}(g) = \mathrm{id}_A \in \mathrm{Aut}(A)$ for every g, and the identity map is trivially a group homomorphism from G into $\mathrm{Aut}(A)$.

2.2.1 Crossed-product C^*-algebra

In this subsection we introduce the notion of a crossed-product C^*-algebra, in the limited form needed for our purposes; for a more general treatment we refer the Reader to Blackadar [2]. The definitions below follow Sierakowski [9].

Definition 2.2.5 (C^*-dynamical system)**.** A *C^*-dynamical system* is a triple (A, G, α), where A is a C^*-algebra, G is a discrete group, and $\alpha\colon G \to \mathrm{Aut}(A)$ is a group action on A. For a C^*-dynamical system (A, G, α) we set

$$C_c(G, A, \alpha) = \{f\colon G \to A : f \text{ has finite support}\},$$

the space of all A-valued functions on G. We usually write $g \cdot a$ for $\alpha(g)(a)$.

If G is finite, every $f : G \to A$ is automatically finitely supported. A typical element $a \in C_c(G, A, \alpha)$ is written $a = \sum_{g \in G} a_g u_g$, where only finitely many of the $a_g \in A$ are nonzero, and $\{u_g\}_{g \in G}$ is a formal family of symbols indexed by G (which specific representation the u_g correspond to depends on the crossed product in question).

The space $C_c(G, A, \alpha)$ carries both a product and an involution, defined precisely so that the group action becomes *inner*, i.e. so that $g \cdot a = u_g a u_g^*$ for all $g \in G$, $a \in A$. Explicitly, for $a = \sum_{g \in G} a_g u_g$ and $b = \sum_{h \in G} b_h u_h$ in $C_c(G, A, \alpha)$, the product and involution are given by

$$ab = \sum_{g,h \in G} a_g\,(g \cdot b_h)\,u_{gh}, \qquad a^* = \sum_{g \in G} (g^{-1} \cdot a_g^*)\,u_{g^{-1}}.$$

Definition 2.2.6 (Covariant representation)**.** Fix a dynamical system (A, G, α). A *covariant representation* $(\pi, u, \mathcal{H})$ of (A, G, α) consists of a unitary representation $u\colon G \to B(\mathcal{H})$ of G and a representation $\pi\colon A \to B(\mathcal{H})$ of A such that

$$u(g)\,\pi(a)\,u(g)^* = \pi(g \cdot a)$$

for every $g \in G$ and $a \in A$.

Definition 2.2.7 (Full C^*-algebra norm on $C_c(G, A, \alpha)$)**.** The *full C^*-algebra norm* on $C_c(G, A, \alpha)$ is given by

$$\|\cdot\| = \sup \|(\pi \times u)(\cdot)\|,$$

where the supremum ranges over all covariant representations $(\pi, u, \mathcal{H})$ of (A, G, α).

Definition 2.2.8 (Crossed product C^*-algebra)**.** The *crossed product* of a C^*-algebra A and a discrete group G is the completion of $C_c(G, A, \alpha)$ with respect to the full norm. We denote it by

$$\tilde{B} \rtimes_{\hat{\alpha}} \hat{G}.$$

Remark 2.2.3. In this work we will not make use of norms of crossed product C^*-algebras and almost all groups will be discrete, so we can treat crossed product C^*-algebras as formal linear combinations of some family $\{u_g\}_{g \in G}$, which is dependent on the group action in question.

The Reader will find some examples of such C^*-algebras below.

Example 2.2.4 (Trivial action recovers the group C^*-algebra). Let $A = \mathbb{C}$ with the trivial action $\alpha_g = \mathrm{id}$ for all $g \in G$. Then every element of $C_c(G, A, \alpha)$ is just a finitely supported function $G \to \mathbb{C}$, with convolution product and involution reducing to the usual group algebra structure, since the action contributes no twisting. Completing in the full norm yields

$$\mathbb{C} \rtimes_\alpha G \;\cong\; C^*(G),$$

the (full) group C^*-algebra of G. For instance, taking $G = \mathbb{Z}$ gives $\mathbb{C} \rtimes \mathbb{Z} \cong C(\mathbb{T})$, since $C^*(\mathbb{Z})$ is generated by a single unitary with no relations, i.e. the continuous functions on the circle via the Fourier transform.

Example 2.2.5 (Irrational rotation algebra). Let $A = C(\mathbb{T})$, the continuous functions on the torus, and let $G = \mathbb{Z}$ act by rotation through angle $2\pi\theta$ for some fixed $\theta \in \mathbb{R} \setminus \mathbb{Q}$, i.e.

$$\alpha_n(f)(z) = f\big(e^{-2\pi i n\theta} z\big), \qquad z \in \mathbb{T},\ n \in \mathbb{Z}.$$

The resulting crossed product

$$A_\theta \;:=\; C(\mathbb{T}) \rtimes_\alpha \mathbb{Z}$$

is the *irrational rotation algebra* (a noncommutative torus). It is generated by a unitary u (implementing the $\mathbb{Z}$-action) and a unitary v (generating $C(\mathbb{T})$) subject to the single relation

$$uv = e^{2\pi i\theta}\, vu,$$

and is a standard first example of a simple, noncommutative C^*-algebra arising from a crossed product.

2.3 Matrix Lie Groups and Algebras

In this section we recall some notions from the theory of matrix Lie groups and algebras that will be needed later in the thesis. A *Lie group* is a group that is simultaneously a smooth manifold, in such a way that both the group multiplication and the inversion map are smooth. A *Lie algebra* $\mathfrak{g}$ is a vector space over a field (for us, always $\mathbb{C}$) equipped with a bilinear form, the *Lie bracket*,

$$[\cdot,\cdot]\colon \mathfrak{g}\times\mathfrak{g}\to\mathfrak{g},$$

which is antisymmetric and satisfies the Jacobi identity:

- $[x, y] = -[y, x]$,
- $[x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0$.

These two notions are intimately related: every Lie group gives rise to a Lie algebra, obtained as the tangent space at the identity element, together with a bracket induced by the group structure.

Our primary interest lies in *matrix Lie groups*, i.e. subgroups of the general linear group

$$GL(n,\mathbb{C}) = \{A \in M_n(\mathbb{C}) : \det(A) \neq 0\},$$

taken with matrix multiplication as the group operation. Important examples include the unitary group

$$U(n) = \{U \in M_n(\mathbb{C}) : U^*U = UU^* = I\},$$

its subgroup, the special unitary group

$$SU(n) = \{U \in M_n(\mathbb{C}) : U^*U = UU^* = I,\ \det(U) = 1\},$$

and the projective unitary group

$$PU(n) = U(n)/Z(U(n)),$$

where $Z(U(n))$ denotes the center of $U(n)$. Each of these groups carries the subspace topology inherited from $M_n(\mathbb{C})$, except for $PU(n)$, which instead carries the quotient topology induced by the projection $U(n) \to U(n)/Z(U(n))$.

Proposition 2.3.1. $Z(U(n)) \cong \mathbb{T}$.

Proof. Recall that the center of $U(n)$ is

$$Z(U(n)) = \{A \in U(n) : AU = UA \text{ for all } U \in U(n)\}.$$

Let $A \in Z(U(n))$. Since A commutes with every unitary matrix, it in particular commutes with every permutation matrix. Let P_{ij} denote the permutation matrix exchanging the i-th and j-th basis vectors. From

$$AP_{ij} = P_{ij}A$$

we see that conjugating A by P_{ij} leaves it unchanged. But conjugation by P_{ij} swaps the i-th and j-th rows and columns of A; since this holds for every pair i, j, all off-diagonal entries of A must vanish and all diagonal entries must coincide. Hence $A = \lambda I_n$ for some $\lambda \in \mathbb{C}$, where by I_n we denote the $n \times n$-identity matrix.

As $A \in U(n)$, we have $A^*A = I$, and substituting $A = \lambda I_n$ gives

$$(\overline{\lambda} I_n)(\lambda I_n) = |\lambda|^2 I_n = I_n,$$

so $|\lambda| = 1$.

Conversely, if $A = \lambda I_n$ with $|\lambda| = 1$, then clearly $A \in U(n)$, and for every $U \in U(n)$,

$$AU = \lambda U = UA,$$

so $A \in Z(U(n))$. This proves the claim. □

We will also make use of

$$SO(n) = \{R \in M_n(\mathbb{R}) : R^T R = RR^T = I,\ \det(R) = 1\},$$

the Lie group of orientation-preserving rotations fixing the origin.

For matrix Lie algebras, the Lie bracket is given by the matrix commutator

$$[X, Y] = XY - YX$$

for X, Y in the Lie algebra; one readily checks that this bracket is antisymmetric and satisfies the Jacobi identity.

The Lie algebra associated to $GL(n, \mathbb{C})$ is denoted $\mathfrak{gl}(n, \mathbb{C})$ and consists of *all* complex $n \times n$ matrices,

$$\mathfrak{gl}(n, \mathbb{C}) = M_n(\mathbb{C}).$$

The Lie algebra associated to $SU(n)$, denoted $\mathfrak{su}(n)$, consists of the traceless skew-Hermitian matrices,

$$\mathfrak{su}(n) = \{X \in M_n(\mathbb{C}) : X^* = -X,\ \mathrm{Tr}(X) = 0\}.$$

We may regard $\mathfrak{su}(2)$ as a 3-dimensional real vector space (why?).

Example 2.3.1 (The circle group $SO(2)$ and its Lie algebra). The group $SO(2)$ consists of rotation matrices

$$R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, \qquad \theta \in [0, 2\pi),$$

and is isomorphic, as a Lie group, to the circle $\mathbb{T}$ via $\theta \mapsto R_\theta$. Differentiating the defining relation $R^T R = I$ at $\theta = 0$ shows that the corresponding Lie algebra $\mathfrak{so}(2)$ consists of antisymmetric real matrices,

$$\mathfrak{so}(2) = \left\{ \begin{pmatrix} 0 & -t \\ t & 0 \end{pmatrix} : t \in \mathbb{R} \right\},$$

a one-dimensional real vector space. Since any two elements of $\mathfrak{so}(2)$ are scalar multiples of one another, the Lie bracket on $\mathfrak{so}(2)$ vanishes identically: it is an *abelian* Lie algebra.

Example 2.3.2 (A basis for $\mathfrak{su}(2)$). A convenient real basis for $\mathfrak{su}(2)$ is given by $\{i\sigma_1, i\sigma_2, i\sigma_3\}$, where $\sigma_1, \sigma_2, \sigma_3$ are the Pauli matrices

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Each $i\sigma_k$ is traceless and skew-Hermitian, so $i\sigma_k \in \mathfrak{su}(2)$, and the three matrices are linearly independent over $\mathbb{R}$; together with $\dim_{\mathbb{R}} SU(2) = 3$ this confirms that $\mathfrak{su}(2)$ is a 3-dimensional real vector space, as anticipated above. The commutation relations

$$[i\sigma_1, i\sigma_2] = -2i\sigma_3, \qquad [i\sigma_2, i\sigma_3] = -2i\sigma_1, \qquad [i\sigma_3, i\sigma_1] = -2i\sigma_2$$

exhibit $\mathfrak{su}(2)$ as (a rescaling of) the cross-product Lie algebra on $\mathbb{R}^3$, and identify $\mathfrak{su}(2)$ with $\mathfrak{so}(3)$ up to isomorphism.

Example 2.3.3 (Non-example: $GL(n, \mathbb{R})$ inside $GL(n, \mathbb{C})$). The real general linear group

$$GL(n, \mathbb{R}) = \{A \in M_n(\mathbb{R}) : \det(A) \neq 0\}$$

is a matrix Lie group in its own right, and sits inside $GL(n, \mathbb{C})$ as a subgroup, but *not* as a complex Lie subgroup: its associated Lie algebra

$$\mathfrak{gl}(n, \mathbb{R}) = M_n(\mathbb{R})$$

is only a *real* vector space, and is not closed under multiplication by i. This illustrates that subgroups of $GL(n, \mathbb{C})$ need not inherit a complex Lie algebra structure, even though $\mathfrak{gl}(n, \mathbb{C})$ itself is a complex vector space.

Chapter 3

Quantum sets and quantum graphs

In this chapter we will introduce the notions of quantum sets and quantum graphs. They were first considered by...

3.1 Quantum sets

Let B be a C^*-algebra equipped with a faithful positive functional ψ. By the Gelfand–Naimark theorem, B can be realized concretely as a closed $*$-subalgebra of the bounded linear operators $B(\mathcal{H})$ on some Hilbert space $\mathcal{H}$. When B is finite-dimensional, this abstract picture simplifies considerably: rather than passing to $\mathcal{H}$, we can identify B with its own Hilbert space completion $L^2(B,\psi)$, by equipping B itself with the inner product induced by ψ,

$$\langle x, y\rangle = \psi(x^*y), \qquad x, y \in B.$$

Since ψ is faithful, this form is positive definite, and since B is already finite-dimensional, it is automatically complete; hence $L^2(B,\psi)$ may simply be taken to be B itself, viewed as a Hilbert space via this inner product.

This Hilbert space structure is the key extra piece of data that allows one to speak of adjoints of maps built from B, and in particular of the adjoint of multiplication. This leads to the central definition of this thesis, taken directly from Schäfer i Tobolski [8].

Definition 3.1.1. (Quantum set, Schäfer i Tobolski [8, Definition 2.1]) A *quantum set* (B,ψ) is a pair consisting of a finite-dimensional C^*-algebra B and a faithful positive functional $\psi\colon B \to \mathbb{C}$ such that

$$mm^* = \mathrm{Id}_B$$

holds, where $m\colon B \otimes B \to B$ is the multiplication map and m^* is its adjoint with respect to the inner products induced by ψ, i.e. the unique linear map determined by

$$\langle m(a \otimes b), c\rangle = \langle a \otimes b, m^*(c)\rangle_{\psi\otimes\psi}$$

for all $a, b, c \in B$.

Remark 3.1.1. We refer to any functional ψ satisfying the condition in the definition above as a *quantum set functional* on B. The condition $mm^* = \mathrm{Id}_B$ should be thought of as a normalization condition on ψ: not every faithful positive functional on a finite-dimensional C^*-algebra makes (B, ψ) into a quantum set, only those compatible with the multiplication in this precise sense.

Remark 3.1.2. Recall that, more generally, a *comultiplication* on a C^*-algebra A is a $*$-homomorphism

$$\Delta\colon A \to A \otimes A$$

making the following diagram commute (coassociativity):

$$\begin{tikzcd} A \arrow[r, "\Delta"] \arrow[d, "\Delta"'] & A \otimes A \arrow[d, "\Delta\otimes\mathrm{id}"] \\ A \otimes A \arrow[r, "\mathrm{id}\otimes\Delta"'] & A \otimes A \otimes A \end{tikzcd}$$

In general such a Δ is far from unique, and need not interact with any given inner product on A at all. In our setting, however, the situation is much more rigid: once (B, ψ) is a quantum set, the adjoint m^* of the multiplication map is automatically a comultiplication on B, and it is the *unique* one arising this way, since it is pinned down by m together with the Riesz representation theorem for the Hilbert space $L^2(B, \psi)$.

Remark 3.1.3. Because in our setup the comultiplication is obtained from the ordinary multiplication m purely by adjunction with respect to ψ, we will not introduce separate notation for it, and will simply write m^* throughout.

Example 3.1.1 (Classical finite sets as quantum sets). Every finite set X gives rise to a quantum set $(C(X), \psi_X)$, where $C(X)$ denotes the C^*-algebra of complex-valued functions on X with pointwise multiplication and involution given by pointwise complex conjugation, $f^*(x) = \overline{f(x)}$, and where

$$\psi_X : C(X) \to \mathbb{C}, \qquad \psi_X(f) = \sum_{x\in X} f(x).$$

To see that ψ_X is positive, note that for any $f \in C(X)$ the product f^*f evaluates pointwise as

$$(f^*f)(x) = \overline{f(x)}\, f(x) = |f(x)|^2,$$

so that

$$\psi_X(f^*f) = \sum_{x\in X} |f(x)|^2 \geq 0.$$

For faithfulness, suppose $\psi_X(f^*f) = 0$ for some $f \in C(X)$, i.e.

$$\sum_{x \in X} |f(x)|^2 = 0.$$

Since each summand $|f(x)|^2$ is non-negative, the sum can vanish only if every term does, so $f(x) = 0$ for all $x \in X$, and hence $f = 0$. Thus ψ_X is faithful.

It remains to check the quantum set condition $mm^* = \mathrm{Id}$. Let $\{e_x\}_{x \in X}$ denote the orthonormal basis of $C(X)$ given by the characteristic functions of points, so that

$$m(f \otimes g) = fg, \qquad m^*(e_x) = e_x \otimes e_x.$$

Evaluating the composite mm^* on a basis element gives

$$(mm^*)(e_x) = m\big(m^*(e_x)\big) = m(e_x \otimes e_x) = e_x e_x = e_x,$$

since e_x is idempotent. As mm^* and Id agree on the basis $\{e_x\}_{x \in X}$, they agree on all of $C(X)$ by linearity, so $mm^* = \mathrm{Id}$, and $(C(X), \psi_X)$ is indeed a quantum set.

Remark 3.1.4. The identification of finite sets and commutative C^*-algebras is inspired by Gelfand's duality theorem, which establishes that every commutative C^*-algebra is isomorphic to the algebra of continuous functions on a compact Hausdorff space. In the finite-dimensional case, this space reduces to a discrete finite set, and the algebra's minimal projections correspond directly to the individual elements of the set. Furthermore, the functional ψ_X serves as a quantum analogue of the counting measure (summing over all elements), while the condition $mm^* = \mathrm{Id}$ ensures the correct normalization and structural compatibility of the multiplication map with this measure, generalizing the discrete Cartesian framework of classical sets to the quantum setting.

Example 3.1.2 (The special case $\mathbb{C}^n$). Taking $X = \{1, \ldots, n\}$ in the previous example recovers the quantum set $(\mathbb{C}^n, \psi_n)$, where ψ_n is determined on the standard basis by $\psi_n(e_i) = 1$ and extends linearly to

$$\psi_n(x) = \sum_{i=1}^{n} x_i, \qquad x \in \mathbb{C}^n.$$

Positivity and faithfulness follow exactly as above: writing $x = \sum_i x_i e_i$ and using orthogonality of the basis,

$$x^* x = \Big(\sum_{i=1}^{n} \overline{x_i}\, e_i\Big)\Big(\sum_{j=1}^{n} x_j e_j\Big) = \sum_{i=1}^{n} \overline{x_i} x_i\, e_i = \sum_{i=1}^{n} |x_i|^2 e_i,$$

so that

$$\psi_n(x^* x) = \sum_{i=1}^{n} |x_i|^2 \psi_n(e_i) = \sum_{i=1}^{n} |x_i|^2 \geq 0,$$

which vanishes precisely when every $x_i = 0$, i.e. when $x = 0$. The quantum set condition again reduces to the computation

$$(mm^*)(e_i) = m(e_i \otimes e_i) = e_i, \qquad i = 1, \ldots, n,$$

which forces $mm^* = \mathrm{Id}$ by linearity.

Example 3.1.3 (Matrix algebras). For any $n \in \mathbb{N}_+$, the pair $(M_n, n\,\mathrm{Tr})$ is a quantum set, where Tr denotes the usual trace on M_n. We will later see that this quantum set arises naturally as a crossed product: writing $\hat{\mathbb{Z}}_n$ for the dual of $\mathbb{Z}_n$ acting on $(\mathbb{C}^n, \psi_n)$ via the standard action, one has

$$(M_n, n\,\mathrm{Tr}) \;\cong\; (\mathbb{C}^n \rtimes \hat{\mathbb{Z}}_n,\; \psi_\rtimes),$$

a fact we prove in 3.1.2. $n\,\mathrm{Tr}$ is chosen instead of simply Tr so that the condition $mm^* = \mathrm{Id}$ is fulfilled.

Lemma 3.1.1 (Comultiplication for matrix algebras)**.** *In the quantum set $(M_n, n\,\mathrm{Tr})$, the comultiplication m^* is given on matrix units by*

$$m^*(e_{ij}) = \sum_{k=1}^{n} e_{ik} \otimes e_{kj},$$

where e_{ij} denotes the matrix with a single 1 in the i-th row and j-th column, extended linearly to all of M_n.

Proof. The proof is a direct computation using two standard facts about matrix units:

1. $e_{ab}e_{cd} = \delta_{bc}\, e_{ad}$;
2. $\langle e_{ij}, e_{kl}\rangle = \mathrm{Tr}(e_{ji}e_{kl}) = \delta_{ik}\,\delta_{jl}$,

where δ_{ij} is the Kronecker delta. In particular, (2) shows that $\{e_{ij}\}_{i,j=1,\ldots,n}$ forms an orthonormal basis of M_n with respect to the inner product induced by $n\,\mathrm{Tr}$.

Fix arbitrary basis vectors e_{ab}, e_{cd}. By (1),

$$m(e_{ab} \otimes e_{cd}) = e_{ab}e_{cd} = \delta_{bc}\, e_{ad},$$

and hence, for any basis vector e_{ij},

$$\big\langle m(e_{ab} \otimes e_{cd}),\, e_{ij}\big\rangle = \delta_{bc}\big\langle e_{ad}, e_{ij}\big\rangle = \delta_{bc}\,\delta_{ai}\,\delta_{dj}.$$

On the other hand, pairing $e_{ab} \otimes e_{cd}$ against the proposed adjoint gives

$$\Big\langle e_{ab} \otimes e_{cd},\, \sum_{k=1}^{n} e_{ik} \otimes e_{kj}\Big\rangle = \sum_{k=1}^{n}\langle e_{ab}, e_{ik}\rangle\langle e_{cd}, e_{kj}\rangle = \sum_{k=1}^{n} \delta_{ai}\delta_{bk}\,\delta_{ck}\delta_{dj} = \delta_{ai}\,\delta_{bc}\,\delta_{dj},$$

using that the sum over k collapses to the single term $k = b = c$. The two expressions coincide, so

$$\big\langle m(e_{ab} \otimes e_{cd}), e_{ij}\big\rangle = \big\langle e_{ab} \otimes e_{cd}, m^*(e_{ij})\big\rangle$$

for all basis vectors, and since $\{e_{ij}\}$ is an orthonormal basis, this identifies m^* as claimed on the whole of M_n by linearity. $\square$

3.1.1 Crossed product quantum sets

Schäfer i Tobolski [8] introduced the notion of a *crossed product quantum set*. Suppose we are given a finite-dimensional C^*-algebra $\tilde{B}$ together with a dual group action $\hat{\alpha}\colon \hat{G} \to \operatorname{Aut}(\tilde{B})$, where G is a finite abelian group. One may then form the crossed product

$$B := \tilde{B} \rtimes_{\hat{\alpha}} \hat{G}.$$

If $\tilde{B}$ is equipped with a faithful functional $\tilde{\psi}$ making $(\tilde{B}, \tilde{\psi})$ a quantum set, one can construct a corresponding faithful functional on B making B into a quantum set as well, as follows.

Definition 3.1.2. (Crossed product quantum set, Schäfer i Tobolski [8, Definition 3.1]) Let $(\tilde{B}, \tilde{\psi})$ be a quantum set, and let $\hat{\alpha}\colon \hat{G} \to \operatorname{Aut}(\tilde{B}, \tilde{\psi})$ be an action of the dual of a finite abelian group G. The *crossed product quantum set* is defined as

$$(\tilde{B}, \tilde{\psi}) \rtimes_{\hat{\alpha}} \hat{G} := \big(\tilde{B} \rtimes_{\hat{\alpha}} \hat{G},\ \psi_{\rtimes}\big),$$

where $\tilde{B} \rtimes_{\hat{\alpha}} \hat{G}$ is the crossed product C^*-algebra, and $\psi_{\rtimes}$ is the functional given by

$$\psi_{\rtimes}\Big(\sum_{\chi \in \hat{G}} b_\chi u_\chi\Big) = n\, \tilde{\psi}(b_1),$$

with $n = |\hat{G}|$.

The proof that this definition is well posed, i.e. that the resulting object is indeed a quantum set, can be found in Schäfer i Tobolski [8, Proposition 3.4].

Example 3.1.4. (Schäfer i Tobolski [8, Example 3.5]) If $\hat{\alpha} = \mathrm{triv}$ is the trivial action, then

$$\tilde{B} \rtimes_{\hat{\alpha}} \hat{G} \ \cong\ \tilde{B} \otimes C(G),$$

via the isomorphism $b_\chi u_\chi \mapsto b_\chi \otimes \tilde{u}_\chi$, where $\tilde{u}_\chi \in C(G)$ denotes the function $g \mapsto \chi(g)$ on G.

Lemma 3.1.2. *Let $\hat{\alpha}\colon \hat{\mathbb{Z}}_n \to \operatorname{Aut}(\mathbb{C}^n)$ be the action given by cyclically shifting coordinates. Then*

$$(M_n, n\operatorname{Tr}) \ \cong\ (\mathbb{C}^n \rtimes_{\hat{\alpha}} \hat{\mathbb{Z}}_n,\ \psi_{\rtimes}).$$

Proof. Identify $\hat{\mathbb{Z}}_n$ with $\mathbb{Z}_n = \{0, 1, \dots, n-1\}$ (with arithmetic taken modulo n), so that a general element of $\mathbb{C}^n \rtimes_{\hat{\alpha}} \hat{\mathbb{Z}}_n$ is written $b = \sum_{k \in \mathbb{Z}_n} b_k u_k$ with $b_k \in \mathbb{C}^n$, and the action is $\hat{\alpha}_k(e_i) = e_{i+k}$ on the standard basis $\{e_i\}_{i \in \mathbb{Z}_n}$ of $\mathbb{C}^n$, as in the previous examples.

We construct an explicit $*$-isomorphism $\pi\colon \mathbb{C}^n \rtimes_{\hat{\alpha}} \hat{\mathbb{Z}}_n \to M_n$ as follows. On $\mathbb{C}^n$, let M_{b_k} denote the diagonal matrix with entries $b_k(0), \dots, b_k(n-1)$, i.e. $M_{b_k} = \sum_i b_k(i)\, e_{ii}$ in the matrix unit notation of the earlier lemma. Let S be the cyclic shift matrix defined by $Se_i = e_{i+1}$ for the standard basis vectors e_i of $\mathbb{C}^n$ (indices mod n), and set

$$\pi(b) = \sum_{k\in\mathbb{Z}_n} M_{b_k}\, S^k, \qquad b = \sum_{k\in\mathbb{Z}_n} b_k u_k.$$

π **is a $*$-homomorphism.** A direct computation shows that $S^k M_f S^{-k} = M_{\hat{\alpha}_k(f)}$ for $f \in \mathbb{C}^n$, which is exactly the covariance relation $u_k b u_k^* = \hat{\alpha}_k(b)$ required of the crossed product. Using this, for $a = \sum_k a_k u_k$ and $b = \sum_l b_l u_l$,

$$\pi(a)\pi(b) = \sum_{k,l} M_{a_k} S^k M_{b_l} S^l = \sum_{k,l} M_{a_k}\, M_{\hat{\alpha}_k(b_l)}\, S^{k+l} = \sum_m M_{\sum_{k+l=m} a_k \hat{\alpha}_k(b_l)}\, S^m = \pi(ab),$$

matching the multiplication rule on $C_c(\hat{\mathbb{Z}}_n, \mathbb{C}^n, \hat{\alpha})$ from the previous subsection, and similarly

$$\pi(a)^* = \sum_k S^{-k} M_{a_k^*} = \sum_k M_{\hat{\alpha}_{-k}(a_k^*)}\, S^{-k} = \pi(a^*),$$

matching the involution formula. So π is a $*$-homomorphism.

π **is bijective.** Write $e_i = \delta_i \in \mathbb{C}^n$ for the standard basis vectors. Since S^k sends $e_j \mapsto e_{j+k}$, the operator $M_{e_i} S^k$ sends $e_{i-k} \mapsto e_i$ and annihilates every other basis vector, i.e.

$$\pi(e_i u_k) = M_{e_i} S^k = e_{i,\, i-k},$$

a matrix unit in the sense of the earlier lemma. As i ranges over $\mathbb{Z}_n$ and k ranges over $\mathbb{Z}_n$, the pair $(i, i-k)$ ranges exactly once over all of $\mathbb{Z}_n \times \mathbb{Z}_n$ (for fixed i, varying k makes $i-k$ range over all residues). Hence $\{\pi(e_i u_k)\}_{i,k\in\mathbb{Z}_n}$ is precisely the full set of matrix units $\{e_{ab}\}_{a,b\in\mathbb{Z}_n}$, which forms a basis of M_n. Since $\{e_i u_k\}_{i,k}$ is a basis of $\mathbb{C}^n \rtimes_{\hat{\alpha}} \hat{\mathbb{Z}}_n$ (of the same dimension n^2) and π carries it bijectively to a basis of M_n, the linear map π is a bijection.

π **intertwines the functionals.** Using $\pi(e_i u_k) = e_{i,i-k}$ and the fact that a matrix unit e_{ab} has trace δ_{ab}, we get

$$\mathrm{Tr}\left(\pi(e_i u_k)\right) = \delta_{i,\, i-k} = \delta_{k,0}.$$

Hence for a general element $b = \sum_k b_k u_k = \sum_{k,i} b_k(i)\, e_i u_k$,

$$\mathrm{Tr}(\pi(b)) = \sum_{k,i} b_k(i)\ \mathrm{Tr}\left(\pi(e_i u_k)\right) = \sum_{k,i} b_k(i)\, \delta_{k,0} = \sum_i b_0(i) = \tilde{\psi}(b_0),$$

where $\tilde{\psi}$ is the quantum set functional $\tilde{\psi}(x) = \sum_i x_i$ on $(\mathbb{C}^n, \tilde{\psi})$ from the earlier example, and b_0 is precisely the coefficient at the trivial character (i.e. $k = 0$) appearing in the definition of $\psi_\rtimes$. Multiplying by $n = |\hat{\mathbb{Z}}_n|$,

$$n\, \mathrm{Tr}(\pi(b)) = n\, \tilde{\psi}(b_0) = \psi_\rtimes(b).$$

Thus π is a $*$-isomorphism $\mathbb{C}^n \rtimes_{\hat{\alpha}} \hat{\mathbb{Z}}_n \to M_n$ satisfying $n\,\mathrm{Tr} \circ \pi = \psi_\rtimes$, which is exactly an isomorphism of quantum sets

$$(\mathbb{C}^n \rtimes_{\hat{\alpha}} \hat{\mathbb{Z}}_n,\ \psi_\rtimes)\ \cong\ (M_n,\ n\,\mathrm{Tr}). \qquad \square$$

3.2 Quantum graphs

A graph is classically defined as a pair (V, E) with V serving as the *set of vertices* and $E \subseteq V^2$ serving as the *set of edges*. Such a graph could be interpreted as a matrix $A = [a_{ij}] \in \{0,1\}^{n \times n}$ with $(v_i, v_j) \in E \Leftrightarrow a_{ij} = 1$; such a representation is often called the *adjacency matrix*. The key property of such matrices is that they are invariant with respect to entrywise multiplication. In turn we can use this property to generalise graphs on sets to graphs on quantum sets.

The following definitions are due to Schäfer i Tobolski [8].

Definition 3.2.1. (Quantum adjacency matrix, Schäfer i Tobolski [8, Definition 2.5]) Let (B, ψ) be a quantum set. A *quantum adjacency matrix* on this quantum set is a linear map $A\colon B \to B$ which is Schur-idempotent and *-preserving, by which we mean that it satisfies

$$m(A \otimes A)m^* = A, \qquad A(b^*) = A(b)^* \quad \text{for all } b \in B.$$

Definition 3.2.2. (Quantum graph, Schäfer i Tobolski [8, Definition 2.5]) A *quantum graph* on (B, ψ) is given by a quantum adjacency matrix on this quantum set.

For all intents and purposes we will identify both notions and simply call an appropriate $A\colon B \to B$ both a quantum adjancenty matrix and a quantum graph.

Remark 3.2.1. There are other equivalent definitions of a quantum graph, namely those based on

- an *edge projection*, i.e. a projection $\tilde{A} \in C(X) \otimes C(X)^{\mathrm{op}}$, see Musto, Reutter i Verdon [7],
-

We will make use of the latter definition later in the thesis.

Definition 3.2.3. (Schäfer i Tobolski [8, Definition 4.6]) Let A be a quantum graph on a quantum set (B, ψ). We call the quantum graph

- *loopfree* if $m(A \otimes \mathrm{Id})m^* = 0$,
- *undirected* if $A = A^*$, where the adjoint is taken with respect to the inner product induced by ψ.

3.3 Voltage and Derived Quantum Graphs

In this section we introduce the notions of voltage and derived quantum graphs. Voltage graphs and their derived graphs were first introduced by Gross in Gross [5]; these notions have since been extended to the setting of quantum graphs by Bjorn and Tobolski in Schäfer i Tobolski [8].

3.3.1 Classical voltage graphs

Before turning to the quantum setting, it is instructive to first recall the classical construction. Informally, a voltage graph is simply a graph whose edges have been labeled, or *weighted*, by elements of a finite group; the derived graph then unfolds this labeled graph into a (typically much larger) covering graph, built by tracking how these group elements accumulate as one walks along paths in the original graph. This construction originates in topological graph theory, where it was used to build graph embeddings on surfaces of higher genus, but it also turns out to provide a rich and flexible source of examples of graphs with interesting symmetry, which is the perspective we adopt here.

Definition 3.3.1 (Classical voltage graph)**.** A *voltage graph* is a pair (Γ, λ) consisting of a (not necessarily simple) finite directed graph Γ, with vertex set V and edge set E, together with a labelling $\lambda\colon E \to G$ of the edges of Γ by elements of a finite group G.

From every voltage graph one can construct a new graph, called its *derived graph*, as follows.

Definition 3.3.2 (Classical derived graph)**.** The *derived graph* Γ^λ of (Γ, λ) has vertex set $V \times G$ and edge set $E \times G$. Whenever $e \in E$ is an edge from u to v in Γ, and $g \in G$ is a group element, the corresponding edge (e, g) in Γ^λ runs from (u, g) to $(v,\, g\lambda(e))$.

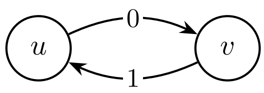
Intuitively, the derived graph consists of $|G|$ labeled copies (*fibers*) of the vertex set of Γ, one for each group element, and every edge e of Γ is lifted to $|G|$ edges in Γ^λ, one leaving each fiber, whose endpoint is determined by translating within the target fiber according to the voltage $\lambda(e)$. This is precisely the picture underlying the worked examples given below.

The following is a major result in the theory of voltage and derived graphs, showing that this construction is not merely a convenient way of producing examples, but in fact accounts for *every* graph admitting a free action of a finite group. It can be found in Gross i Tucker [4, Theorem 2.2.2].

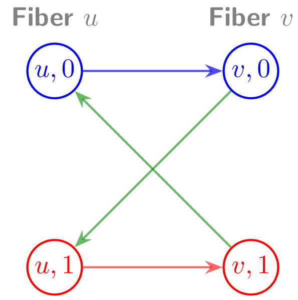
Theorem 3.3.1 (Gross and Tucker theorem)**.** *Let $\tilde{\Gamma}$ be a graph and let G be a finite group acting freely on $\tilde{\Gamma}$ by automorphisms. Then there exists a voltage graph (Γ, λ), with edges labeled by elements of G, such that $\tilde{\Gamma}$ is isomorphic to the derived graph of (Γ, λ). The graph Γ is the quotient graph $\tilde{\Gamma}/G$.*

In other words, voltage graphs classify free group actions on graphs up to isomorphism: recovering (Γ, λ) from $\tilde{\Gamma}$ amounts to nothing more than quotienting by the group action, while reconstructing $\tilde{\Gamma}$ from (Γ, λ) is exactly the derived graph construction above. Below we give some examples of voltage graphs together with their derived graphs.

Directed base graph over $\mathbb{Z}_2$

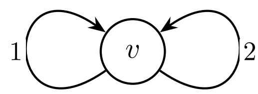


Base voltage graph Γ over $\mathbb{Z}_2$

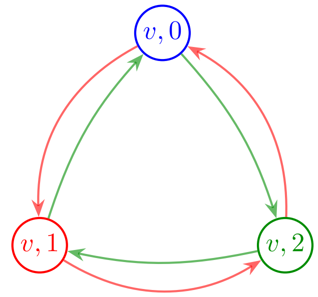


Derived graph Γ^λ (directed 4-cycle)

Directed base graph over $\mathbb{Z}_3$ (single vertex, two loops)

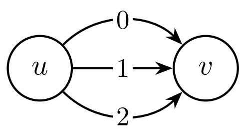


Base voltage graph Γ over $\mathbb{Z}_3$

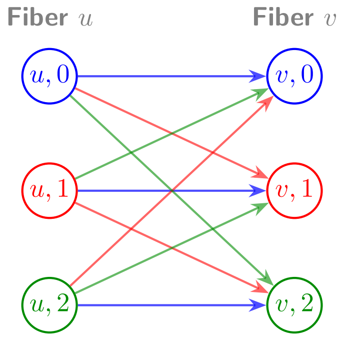


Derived graph Γ^λ (two 3-cycles)

Directed theta graph over $\mathbb{Z}_3$



Base theta graph Γ with edge labels



Derived graph Γ^λ ($K_{3,3}$ colored by edge class)

3.3.2 Generalisation to quantum voltage and derived quantum graphs

We now turn to generalising the Gross–Tucker construction to the setting of quantum graphs. This generalisation was first carried out by Bjorn and Tobolski in Schäfer i Tobolski [8], building on the notion of a crossed product quantum set introduced earlier in this chapter. Just as a classical voltage graph is a labelling of the edges of a graph by group elements, a voltage *quantum* graph will be a family of quantum adjacency matrices indexed by the group G rather than a single labelling function; the derived quantum graph then reassembles this family into a single quantum adjacency matrix on the crossed product quantum set $(B, \psi_{\rtimes})$, playing the role of the derived graph Γ^{λ} from the classical picture. The following definition makes this precise.

Definition 3.3.3. (Schäfer i Tobolski [8, Definition 4.1]) Let $(\tilde{B}, \tilde{\psi})$, $\hat{\alpha}\colon \hat{G} \to \mathrm{Aut}(\tilde{B}, \tilde{\psi})$, $(B, \psi_{\rtimes})$, and $(u_{\chi})_{\chi \in \hat{G}} \subset B$ be as above.

1. For every $g \in G$, define

$$X_g := \frac{1}{n} \sum_{\chi \in \hat{G}} \overline{\chi(g)}\, u_{\chi} u_{\chi}^{\dagger},$$

i.e. X_g is the operator

$$X_g\colon B \to B, \qquad b \mapsto \frac{1}{n} \sum_{\chi \in \hat{G}} \overline{\chi(g)}\, \langle u_{\chi}, b \rangle_{\psi_{\rtimes}}\, u_{\chi}.$$

2. A *voltage quantum graph* on the quantum set $(\tilde{B}, \tilde{\psi})$, with respect to the group action $\hat{\alpha}\colon \hat{G} \to \mathrm{Aut}(\tilde{B}, \tilde{\psi})$, is a family $\tilde{A} = (\tilde{A}_g)_{g \in G}$ of quantum adjacency matrices, indexed by G, satisfying the equivariance condition

$$\tilde{A}_g \hat{\alpha}_{\chi} = \hat{\alpha}_{\chi} \tilde{A}_g$$

for all $g \in G$ and $\chi \in \hat{G}$.

3. The *derived quantum graph* $\tilde{A} \rtimes_{\hat{\alpha}} \hat{G}\colon B \to B$ on $(B, \psi_{\rtimes})$ is then defined by

$$\tilde{A} \rtimes_{\hat{\alpha}} \hat{G} := \sum_{g \in G} m\big(X_g \otimes E_{\rtimes}^{*} \tilde{A}_g E_{\rtimes}\big)\, m^{*},$$

where m denotes the multiplication map on B, i.e. $m = m_{\rtimes}$.

The role played by X_g here is directly analogous to the indicator function of the fiber $V \times \{g\}$ in the classical derived graph: it isolates, within B, the "copy of g" sitting inside the crossed product. The fact that X_g is itself an adjacency matrix on the quantum set $(B, \psi_{\rtimes})$ is checked in Schäfer i Tobolski [8, Proposition 4.5], and the derived quantum graph is then built by combining these fiber projections with the individual voltage components $\tilde{A}_g$, summed over all $g \in G$ – mirroring how, in the classical construction, each edge of Γ^{λ} is assigned both to a specific fiber and to a specific lift of an edge of Γ.

As in the classical case, it will be useful to know when a voltage quantum graph gives rise to a derived quantum graph with no loops, or one that is undirected.

Definition 3.3.4. A voltage quantum graph $\tilde{A} = \{\tilde{A}_g\}_{g \in G}$ is called *loopfree* or *directed* if the respective property from Definition 3.2.3 holds for the sum $\sum_{g \in G} \tilde{A}_g$.

Lemma 3.3.1. *If $A, B \in M_n(\{0,1\})$ and $m^* : \mathbb{C}^n \to \mathbb{C}^n \otimes \mathbb{C}^n$ is the adjoint of the multiplication map m, then $m(A \otimes B)m^* = A \odot B$, the entrywise (Hadamard) product of A and B.*

Proof. Let $\{e_1, \dots, e_n\}$ be the standard basis of $\mathbb{C}^n$. The componentwise multiplication map m acts as:

$$m(e_i \otimes e_j) = \delta_{ij} e_i$$

By definition of the adjoint via the standard inner product, $\langle m(e_i \otimes e_j), e_k \rangle = \langle e_i \otimes e_j, m^*(e_k) \rangle$. Since the left-hand side is $\delta_{ij}\delta_{ik}$, it follows that:

$$m^*(e_k) = e_k \otimes e_k$$

We expand the action of the matrices A and B on a basis vector e_k via their entries A_{ik}, B_{jk}:

$$Ae_k = \sum_{i=1}^{n} A_{ik} e_i, \quad Be_k = \sum_{j=1}^{n} B_{jk} e_j$$

Applying the composite operator $m(A \otimes B)m^*$ to e_k yields:

$$\begin{aligned} m(A \otimes B)m^*(e_k) &= m(Ae_k \otimes Be_k) \\ &= m\left(\sum_{i=1}^{n} \sum_{j=1}^{n} A_{ik} B_{jk} (e_i \otimes e_j) \right) \\ &= \sum_{i=1}^{n} \sum_{j=1}^{n} A_{ik} B_{jk} \delta_{ij} e_i \\ &= \sum_{i=1}^{n} (A_{ik} B_{ik}) e_i \end{aligned}$$

Thus, the (i, k)-th entry of the matrix $m(A \otimes B)m^*$ is exactly $A_{ik} B_{ik}$. Because A and B are binary, this entry evaluates to 1 if and only if $A_{ik} = 1$ and $B_{ik} = 1$, and 0 otherwise. □

Remark 3.3.1. $m(A \otimes B)m^*$ can be thought of as taking the intersecion of two quantum matrices blabla...

3.4 Isomorphisms and Quantum Isomorphisms of Quantum Sets

Quantum isomorphisms were first conveived in Atserias i in. [1] in the context of game theory. We will follow the definitions due to Schäfer i Tobolski [8], which in turn follow the work of Musto, Reutter i Verdon [7].

Definition 3.4.1. (Quantum isomorphisms, Schäfer i Tobolski [8, Definition 2.8]) Let (B_1, ψ_1) and (B_2, ψ_2) be quantum sets, and let $A_1 \colon B_1 \to B_1$ and $A_2 \colon B_2 \to B_2$ be quantum graphs on (B_1, ψ_1) and (B_2, ψ_2), respectively.

- The quantum sets (B_1, ψ_1) and (B_2, ψ_2) are called *isomorphic* if there exists a $*$-isomorphism $\phi \colon B_1 \to B_2$ satisfying $\psi_2 \circ \phi = \psi_1$. The group of all such self-isomorphisms of a quantum set (B, ψ) is denoted $\mathrm{Aut}(B, \psi)$.

- Two quantum graphs are called *isomorphic* if the underlying quantum sets admit an isomorphism $\phi \colon B_1 \to B_2$ which additionally intertwines the graphs, i.e. satisfies $\phi A_1 = A_2 \phi$. The corresponding group of automorphisms of a quantum graph A is written $\mathrm{Aut}(A)$.

- The quantum sets (B_1, ψ_1) and (B_2, ψ_2) are said to be *quantum isomorphic*, denoted $(B_1, \psi_1) \cong_q (B_2, \psi_2)$, if there exist a finite-dimensional Hilbert space $\mathcal{H}$ together with a unital $*$-homomorphism
$$\rho \colon B_1 \to B_2 \otimes B(\mathcal{H})$$
for which the induced map
$$p \colon L^2(B_1) \otimes \mathcal{H} \to L^2(B_2) \otimes \mathcal{H}, \qquad a \otimes \xi \longmapsto \rho(a)(1 \otimes \xi),$$
is a unitary operator between Hilbert spaces.

- Two quantum graphs A_1, A_2 are *quantum isomorphic*, written $A_1 \cong_q A_2$, if the underlying quantum sets admit a quantum isomorphism $\rho \colon B_1 \to B_2 \otimes B(\mathcal{H})$ that in addition satisfies the intertwining condition
$$\rho A_1 = (A_2 \otimes \mathrm{Id}_{B(\mathcal{H})})\, \rho.$$

Chapter 4

Quantum graphs over 3×3 matrices

The classification of quantum graphs on 2×2 matrices was carried out by Matsuda [6] and Gromada [3] independently.

4.1 Pauli matrices and their generalisations

4.1.1 Pauli matrices

An important basis of the Lie algebra $\mathfrak{sl}(2, \mathbb{C})$ is given by the Pauli matrices

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The Pauli matrices are Hermitian, traceless, and linearly independent; since $\dim_{\mathbb{C}} \mathfrak{sl}(2, \mathbb{C}) = 3$, they form a basis of $\mathfrak{sl}(2, \mathbb{C})$.

Because the Pauli matrices are Hermitian rather than skew-Hermitian, they do not themselves belong to $\mathfrak{su}(2)$. Instead, it is the matrices

$$i\sigma_1, \qquad i\sigma_2, \qquad i\sigma_3$$

that form a basis of $\mathfrak{su}(2)$, being both traceless and skew-Hermitian.

4.1.2 Generalised Pauli matrices

In dimensions greater than two, the role of the Pauli matrices is played by the *generalised Pauli matrices*, also known as the *clock* and *shift* matrices. In dimension three, they are defined by

$$P = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \omega & 0 \\ 0 & 0 & \omega^2 \end{pmatrix}, \qquad Q = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix},$$

where $\omega = \exp(2\pi i/3)$.

The matrices P and Q are unitary and satisfy

$$P^3 = Q^3 = I,$$

together with the commutation relation

$$PQ = \omega\, QP.$$

The clock and shift matrices are related by the discrete Fourier transform. The Fourier matrix is

$$F = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{pmatrix},$$

which is unitary, i.e. satisfies $F^*F = FF^* = I$, and exchanges the roles of the clock and shift operators according to

$$FPF^* = Q^{-1}, \qquad FQF^* = P.$$

Consequently, P and Q are unitarily equivalent and represent two complementary bases of the three-dimensional Hilbert space. This duality is the finite-dimensional analogue of the relation between position and momentum operators, and is a fundamental feature of the generalised Pauli matrices.

Lemma 4.1.1. *The matrices P^aQ^b, $a, b \in \{0, 1, 2\}$, form an orthogonal basis of $M_3(\mathbb{C})$ with respect to the Hilbert–Schmidt inner product $\langle A, B\rangle = \mathrm{Tr}(A^*B)$.*

Proof. There are $3^2 = 9$ matrices of the form P^aQ^b, matching $\dim_{\mathbb{C}} M_3(\mathbb{C}) = 9$; it therefore suffices to show that they are pairwise orthogonal.

Since $P^* = P^{-1}$ and $Q^* = Q^{-1}$, we have $(P^aQ^b)^* = Q^{-b}P^{-a}$, so

$$\langle P^aQ^b, P^cQ^d\rangle = \mathrm{Tr}\left(Q^{-b}P^{-a}P^cQ^d\right) = \mathrm{Tr}\left(Q^{-b}P^{c-a}Q^d\right).$$

Using the commutation relation $Q^kP^\ell = \omega^{-k\ell}P^\ell Q^k$, we obtain

$$Q^{-b}P^{c-a} = \omega^{b(c-a)}\, P^{c-a}Q^{-b},$$

and hence

$$\langle P^aQ^b, P^cQ^d\rangle = \omega^{b(c-a)}\,\mathrm{Tr}\left(P^{c-a}Q^{d-b}\right).$$

If $b \neq d$, then Q^{d-b} is a nontrivial permutation matrix, so $P^{c-a}Q^{d-b}$ has zero diagonal entries and $\mathrm{Tr}(P^{c-a}Q^{d-b}) = 0$.

If $b = d$, then

$$\langle P^a Q^b, P^c Q^b \rangle = \mathrm{Tr}(P^{c-a}).$$

Since $P^{c-a} = \mathrm{diag}(1, \omega^{c-a}, \omega^{2(c-a)})$,

$$\mathrm{Tr}(P^{c-a}) = 1 + \omega^{c-a} + \omega^{2(c-a)} = \begin{cases} 3, & a = c, \\ 0, & a \neq c, \end{cases}$$

using $1 + \omega + \omega^2 = 0$.

Combining both cases gives

$$\langle P^a Q^b, P^c Q^d \rangle = 3\, \delta_{ac} \delta_{bd},$$

so the nine matrices $P^a Q^b$ are pairwise orthogonal, and therefore form an orthogonal basis of $M_3(\mathbb{C})$. $\square$

4.2 Gromada's theorem

In this section we state...

The following theorem is due to Gromada [3].

Theorem 4.2.1 (Gromada [3], Theorem 3.7). *Let $m \in \{0, 1, \ldots, n^2\}$. Quantum graphs on M_n with m quantum edges are in one-to-one correspondence with m-dimensional subspaces $V \subseteq \mathfrak{gl}_n$, the adjacency matrix being given by $A_V = \sum_{s=1}^m \xi_s \otimes \xi_s^*$, where $(\xi_1, \ldots, \xi_m)$ is any orthonormal basis of V.*

The quantum graph has no loops if and only if $V \subseteq \mathfrak{sl}_n$, and is undirected if and only if $V = V^*$.

If $V, W \subseteq \mathfrak{gl}_n$ correspond to isomorphic quantum graphs, then $V = UWU^*$ for some $U \in SU_3$.

Proof. See Gromada's paper for the proof. $\square$

Lemma 4.2.1 (Gromada [3], Lemma 3.8). *Every vector subspace $V \subseteq \mathfrak{gl}_n$ with $V = V^*$ admits a self-adjoint basis $\{\xi_s\}$, i.e. with $\xi_s = \xi_s^*$ for every s.*

Proof. We construct the desired basis inductively. Suppose we have already obtained a linearly independent set of self-adjoint vectors $\{\xi_s\}$ spanning a subspace $V_1 \subseteq V$. By construction, V_1 is invariant under the involution, i.e. $V_1 = V_1^*$.

If $V_1 \neq V$, let $V_2 = V_1^{\perp}$ be the orthogonal complement of V_1 with respect to the inner product. Since the inner product and the involution are compatible, V_2 is also invariant under $\cdot^*$.

Choose any nonzero $\xi \in V_2$ and decompose it into its self-adjoint real and imaginary parts,

$$\xi_{\mathrm{Re}} = \frac{\xi + \xi^*}{2}, \qquad \xi_{\mathrm{Im}} = \frac{i(\xi^* - \xi)}{2}.$$

Both ξ_{Re} and ξ_{Im} are self-adjoint elements of V_2, and since $\xi \neq 0$, at least one of them is nonzero. We extend our independent set by appending either both (if linearly independent) or the single nonzero one (if not). Iterating this process eventually produces a full self-adjoint basis of V. $\square$

Remark 4.2.1. The following lemma shows that any undirected graph on M_n may be thought of as being built out of symmetric quantum edges alone.

Lemma 4.2.2. *A complex vector space $V = V^* \subseteq \mathfrak{gl}_n$ uniquely determines a real vector space, spanned by a basis $\{\xi_s\}$ of self-adjoint elements $\xi_s = \xi_s^*$.*

Proof. Let $V \subseteq \mathfrak{gl}_n$ be a complex vector space with $V = V^*$.

Since V is invariant under the Hermitian adjoint, every $v \in V$ admits the decomposition

$$v = \frac{v + v^*}{2} + i\,\frac{v - v^*}{2i}.$$

Both summands lie in V, as V is closed under addition, scalar multiplication, and the adjoint operation, and moreover

$$\left(\frac{v + v^*}{2}\right)^* = \frac{v + v^*}{2}, \qquad \left(i\,\frac{v - v^*}{2i}\right)^* = i\,\frac{v - v^*}{2i},$$

so both are self-adjoint.

Now let $\xi \in V$ be self-adjoint, and let $\{\xi_s\}$ be a self-adjoint basis of V. Write $\xi = \sum_s c_s \xi_s$ for some $c_s \in \mathbb{C}$. Taking the Hermitian adjoint gives

$$\xi = \xi^* = \sum_s \overline{c_s}\,\xi_s,$$

since each ξ_s is self-adjoint, and by uniqueness of the expansion with respect to $\{\xi_s\}$ we get $c_s = \overline{c_s}$ for every s, i.e. every coefficient c_s is real.

Hence every self-adjoint element of V is a real linear combination of $\{\xi_s\}$, so the self-adjoint part of V is a real vector space uniquely determined by V. $\square$

By the above lemma, we may regard any self-adjoint $V \subseteq \mathfrak{sl}_n$ as a real subspace of $\mathfrak{sl}_n$.

Remark 4.2.2. A self-adjoint subspace $V \subseteq \mathfrak{sl}_3$ generated by the generalised Pauli matrices can only have even dimension, i.e. $\dim(V) = m \in \{0, 2, 4, 6, 8\}$. Indeed, the non-identity Pauli basis generators $P^a Q^b \in \mathcal{B}$ are non-self-adjoint (as $(a, b) \neq (3 - a, 3 - b)$ in $\mathbb{Z}_3 \times \mathbb{Z}_3$) and split into mutually disjoint adjoint pairs $\{v, v^*\}$. Since $V = V^*$, including any generator $v \in V$ forces the inclusion of its linearly independent partner v^*, so V decomposes as a direct sum of 2-dimensional reflection-invariant blocks $\mathcal{P}_k = \operatorname{span}\{v, v^*\}$. Thus $\dim(V) = 2r$ for some integer $r \geq 0$.

To characterize all possible loop-free, undirected quantum graphs, it therefore suffices to focus on these relevant subspaces.

4.3 Classification of self-reflective subspaces generated by P and Q

4.3.1 Fundamental reflection-invariant pairs

Since P and Q are unitary ($P^* = P^2$, $Q^* = Q^2$), the conjugate transpose of an arbitrary generator satisfies

$$(P^aQ^b)^* = \omega^{-ab}\, P^{3-a}Q^{3-b}.$$

Because any self-reflective subspace must be closed under conjugate transposition, each basis generator P^aQ^b must appear alongside its structural reflection partner $P^{3-a}Q^{3-b}$. This partitions the eight non-identity generators into four mutually exclusive, reflection-invariant blocks of dimension two:

$$\begin{aligned}
\mathcal{P}_1 &= \{P, P^2\} && \text{(purely diagonal block)}\\
\mathcal{P}_2 &= \{Q, Q^2\} && \text{(purely shift block)}\\
\mathcal{P}_3 &= \{PQ, P^2Q^2\} && \text{(first mixed block)}\\
\mathcal{P}_4 &= \{P^2Q, PQ^2\} && \text{(second mixed block)}
\end{aligned}$$

4.3.2 Action of the discrete Fourier transform

Let F denote the 3×3 discrete Fourier transform matrix. The unitary similarity transformation $X \mapsto FXF^*$ acts as an automorphism of the generalised Pauli algebra, exchanging clock and shift operators:

$$FPF^* = Q^2, \qquad FQF^* = P.$$

On the fundamental blocks $\mathcal{P}_k$, this induces a natural pair-swapping involution:

$$F\mathcal{P}_1F^* = \mathcal{P}_2, \qquad F\mathcal{P}_2F^* = \mathcal{P}_1, \qquad F\mathcal{P}_3F^* = \mathcal{P}_4, \qquad F\mathcal{P}_4F^* = \mathcal{P}_3.$$

4.3.3 Subspace classification theorem

Because the fundamental building blocks $\mathcal{P}_k$ are non-overlapping 2-dimensional sets, every self-reflective subspace generated by powers of P and Q is formed by taking a direct sum of a subset of these blocks (including the empty set).

Theorem 4.3.1. *Every self-reflective subspace $V \subseteq M_3$ generated by P and Q is classified by its dimension $m = \dim(V)$, as recorded in Table 4.1.*

m	Subspaces	Count
$1, 3, 5, 7$	none exist	0
0	$V_0 = \{0\}$	1
2	$V_k = \mathrm{span}(\mathcal{P}_k),\ k \in \{1,2,3,4\}$	4
4	$W_{i,j} = \mathrm{span}(\mathcal{P}_i \oplus \mathcal{P}_j),\ 1 \le i < j \le 4$	6
6	$U_k = \mathrm{span}\left(\bigoplus_{i \neq k} \mathcal{P}_i\right),\ k \in \{1,2,3,4\}$	4
8	$\mathfrak{su}_3 = \mathrm{span}\left(\bigoplus_{k=1}^{4} \mathcal{P}_k\right)$	1

Table 4.1: Classification of self-reflective subspaces $V \subseteq M_3$ generated by P and Q, by dimension.

Corollary 4.3.1. Including the trivial subspace V_0, there exist precisely 16 self-reflective subspaces generated by P and Q.

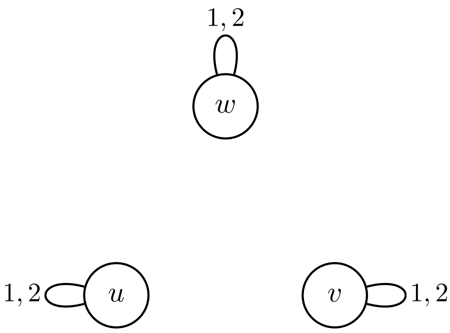
The classification above reduces to elementary combinatorics once the blocks $\mathcal{P}_1, \ldots, \mathcal{P}_4$ are fixed: because these four blocks partition the eight non-identity Pauli generators into disjoint reflection-invariant pairs, choosing a self-reflective subspace V amounts to nothing more than choosing a subset $S \subseteq \{1,2,3,4\}$ of blocks to include, with $V = \bigoplus_{k \in S} \mathcal{P}_k$. There are $2^4 = 16$ such subsets in total, matching the count in the corollary, and since each included block contributes exactly 2 dimensions, the resulting dimension is always even, explaining why V can never have odd dimension. The counts in each row of the table are then just binomial coefficients $\binom{4}{r}$ for $r = |S|$ blocks chosen (namely $1, 4, 6, 4, 1$ for $r = 0, 1, 2, 3, 4$), which is also why the table is symmetric under $m \leftrightarrow 8 - m$: choosing r blocks to include is the same data as choosing $4 - r$ blocks to exclude.

4.4 Classification of loopfree and undirected graphs on M_3

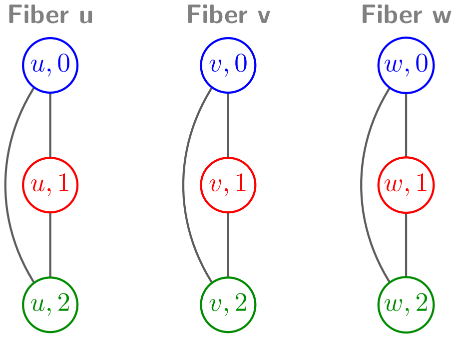
In this section we exhibit, for each of the sixteen self-reflective subspaces classified above, the corresponding voltage graph on $\mathbb{Z}_3$ together with its derived (covering) graph. Four representative cases are shown below; the base graph Γ has its edges and loops labeled by elements of $\mathbb{Z}_3$, and the derived graph Γ^λ consists of three fibers of three vertices each, with edge colors indicating voltage:

— voltage 0 — voltage 1 — voltage 2

Case 1 of 4: Only loops labeled $1, 2$

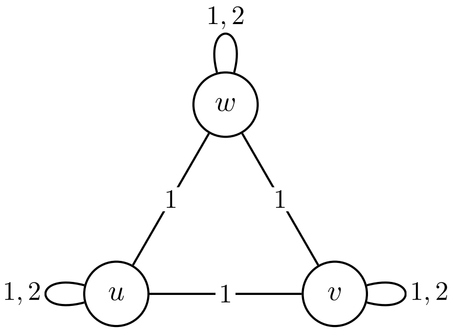


Base Γ: undirected loops

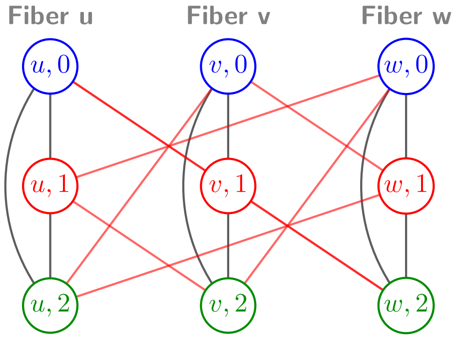


Derived Γ^λ: undirected 3-cycles

Case 2 of 4: Loops plus a 1-voltage triangle

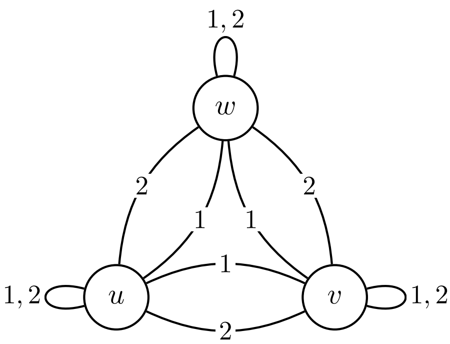


Base Γ: undirected triangle & loops

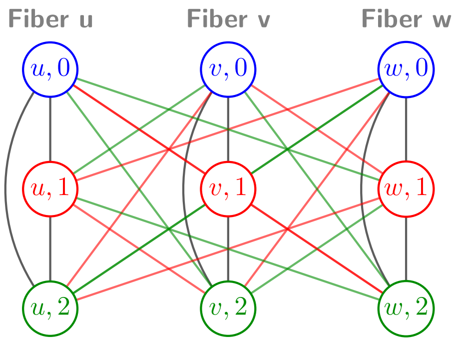


Derived Γ^λ: exactly 18 undirected edges

Case 3 of 4: Loops plus 1- and 2-voltage triangle edges

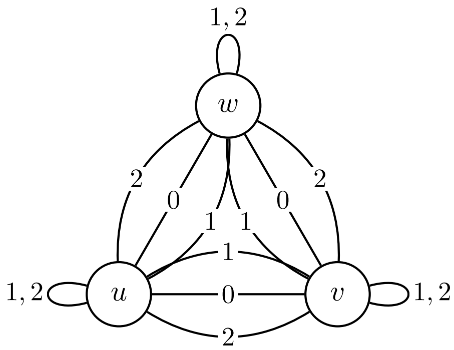


Base Γ: undirected multigraph

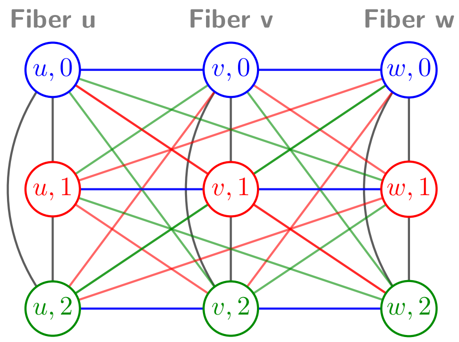


Derived Γ^λ: partial $K_{3,3}$ between fibers

Case 4 of 4: Loops plus 0-, 1-, and 2-voltage triangle edges



Base Γ: full undirected multigraph



Derived Γ^λ: perfect $K_{3,3}$ between all fibers

4.5 Classification of different types of graphs

In this section we consider the prospects for classifying quantum graphs that are not necessarily loop-free and/or undirected.

Theorem 4.5.1. *(Gromada [3, Theorem 3.11]) A simple graph on M_2 is determined up to isomorphism solely by the number of quantum edges $m \in \{0, 1, 2, 3\}$.*

The proof of this theorem relies on the well-known isomorphism $PU(2) \cong SO(3)$, which plays a key role throughout. This result does not generalise to higher dimensions, as the following lemma shows.

Lemma 4.5.1. *The natural action*

$$PU(3) \curvearrowright \mathbb{P}(\mathbb{R}^8) = \{W \leq \mathbb{R}^8 : \dim(W) = 1\}$$

has infinitely many orbits.

Proof. Let $V = \mathfrak{su}(3) \cong \mathbb{R}^8$, and identify $\mathbb{P}(\mathbb{R}^8)$ with $\mathbb{P}(V)$. The action of $PU(3)$ on V is the adjoint representation

$$g \cdot X = gXg^{-1},$$

which induces an action on the 1-dimensional subspaces of V. To show that this action has infinitely many orbits, it suffices to exhibit a projective invariant taking infinitely many values.

Define

$$q(X) = -\operatorname{Tr}(X^2), \qquad c(X) = i \operatorname{Tr}(X^3).$$

Since the trace is invariant under conjugation,

$$q(gXg^{-1}) = q(X), \qquad c(gXg^{-1}) = c(X)$$

for every $g \in PU(3)$. Moreover, for every $\lambda \in \mathbb{R}$,

$$q(\lambda X) = \lambda^2 q(X), \qquad c(\lambda X) = \lambda^3 c(X),$$

so the ratio

$$I(X) = \frac{c(X)^2}{q(X)^3}$$

is invariant both under the adjoint action and under rescaling by nonzero scalars, and hence defines a well-defined invariant of the induced action on $\mathbb{P}(V)$.

Consider now the family

$$X_a = i\,\mathrm{diag}(a, 1, -a-1), \qquad a \in \mathbb{R}.$$

A direct computation gives

$$q(X_a) = 2(a^2 + a + 1), \qquad c(X_a) = -3a(a+1),$$

and therefore

$$I(X_a) = \frac{9a^2(a+1)^2}{8(a^2+a+1)^3}.$$

This is a nonconstant continuous function of a; for instance,

$$I(X_0) = 0, \qquad I(X_1) = \frac{1}{6},$$

so $I(X_a)$ takes infinitely many values as a ranges over $\mathbb{R}$. Since I is constant on each $PU(3)$-orbit of $\mathbb{P}(V)$, distinct values of I must correspond to distinct orbits. Hence the action of $PU(3)$ on $\mathbb{P}(\mathbb{R}^8)$ has infinitely many orbits. □

4.5.1 Simple quantum graph that is not quantum isomprphic to a classical graph

Example 4.5.1. (Gromada [3, Example 8.5]) The matrix

$$\lambda_8 = \frac{1}{\sqrt{2}} \begin{matrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{matrix},$$

which is one of the *Gell–Mann matrices* spanning the Lie algebra $\mathfrak{su}_3$. Then the adjacency matrix $A = \lambda_8 \otimes \lambda_8$ defines a simple graph on M_3.

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